



StayVerse

Submitted:

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1. Project Description

Overview

The proposed project is a full-stack MERN-based accommodation booking platform similar to Airbnb that enables users to **discover, list, and book short-term rentals such as apartments, houses, and vacation homes**. The platform provides a seamless experience for both travelers searching for accommodations and hosts who want to rent out their properties.

The system will feature modern SaaS-style dashboards, AI-powered recommendations, analytics, secure authentication, booking management, and responsive design, ensuring usability across desktops, tablets, and mobile devices.

Target Users

The platform is designed for two main types of users:

Travelers

- People looking for short-term accommodation
- Students or professionals traveling for work
- Tourists planning trips

Hosts

- Property owners who want to rent their homes or apartments
- Individuals offering short-term rentals

Purpose and Need

With the growth of the sharing economy and digital tourism, users increasingly rely on online platforms to find accommodations. Traditional hotel booking systems often lack personalized recommendations, flexible listings, and peer-based hosting models.

This project addresses these needs by providing:

- A user-friendly property discovery system
- AI-powered personalized accommodation recommendations
- Secure booking and payment management
- Host dashboards with analytics
- Property management tools

The platform will simulate the functionality of real-world travel marketplaces while showcasing modern full-stack development practices.

2. Functional Requirements

Below are the functional requirements of the system.

2.1 User Authentication System

Description

The platform will provide a secure authentication system that allows users to register, log in, and manage their accounts.

Objectives

- Enable users to create accounts using email and password
- Secure authentication using **JWT tokens**
- Password encryption using **bcrypt**
- Session management and protected routes
- Role-based access (Guest / Host / Admin)

Features

- User registration
- Login / logout
- Password reset
- Secure token authentication

2.2 Dashboard with Navbar and Sidebar

Description

The system will provide a **centralized dashboard interface** with a navbar and sidebar for easy navigation.

Objectives

- Provide quick access to all features
- Improve usability and organization

Features

- Booking management
- Property listings
- Profile management
- Notifications
- Analytics overview

2.3 Responsive Multi-Device Optimization

Description

The platform will support **desktop, tablet, and mobile devices**.

Objectives

- Ensure accessibility on all screen sizes
- Improve user experience

Features

- Responsive UI using **TailwindCSS**
- Mobile-friendly navigation
- Adaptive layouts

2.4 Property Listings Management (CRUD)

Description

Hosts can create, update, and manage property listings.

Objectives

- Enable hosts to manage rental properties easily

Features

- Add property
- Edit property
- Delete property
- Upload property images
- Set price and availability

2.5 User Profiles

Description

Each user will have a personal profile displaying relevant information.

Objectives

- Build trust and identity within the platform

Features

- Profile picture
- Biography
- Contact information
- Social media links
- Booking history

2.6 Property Search and Filtering

Description

Users can search for accommodations using multiple filters.

Objectives

- Improve property discovery

Features

- Search by location
- Price range filter
- Property type filter
- Availability filter

2.7 Google Maps Location Integration

Description

Properties will be displayed on **Google Maps**.

Objectives

- Help users visualize property locations

Features

- Map-based search
- Property location markers
- Nearby places visualization

2.8 AI Recommendation System

Description

The system will recommend properties based on user behavior and preferences.

Objectives

- Improve user experience with personalized suggestions

Features

- AI-based property suggestions
- Similar listing recommendations
- Location-based recommendations

2.9 Booking Management System

Description

Users can book properties and manage reservations.

Objectives

- Provide a seamless booking experience

Features

- Booking request
- Booking confirmation
- Booking cancellation

- Booking history

2.10 Notifications System

Description

Users will receive notifications about bookings and platform updates.

Objectives

- Keep users informed in real time

Features

- Booking confirmations
- Reservation updates
- Host responses

2.11 Analytics Dashboard

Description

Hosts will be able to monitor property performance.

Objectives

- Provide insights into listing performance

Features

- Booking statistics
- Revenue analytics
- Occupancy rate charts

2.12 PDF Generation

Description

The platform will generate **booking receipts and reservation confirmations**.

Objectives

- Provide official documentation

Features

- Booking receipt generation
- Reservation confirmation PDF

2.13 Blog System

Description

The platform will include a blog section related to travel and accommodations.

Objectives

- Improve engagement and SEO

Features

- Blog creation
- Blog editing
- Blog categories
- Blog search

2.14 Social Media Integration

Description

Users can share properties through social media.

Objectives

- Increase platform visibility

Features

- Share property links
- Social media buttons

2.15 Third-Party API Integration

Description

External services will enhance platform functionality.

APIs

- **Google Maps API** – property location
- **Gemini API** – recommendation system
- **Payfast (test mode)** – simulated payments

2.16 Security and Error Handling

Description

Security measures will protect user data and platform operations.

Objectives

- Ensure safe and reliable system operation

Features

- Password encryption
- Input validation
- API rate limiting
- Error logging

3. Conclusion

The proposed project aims to develop a modern and fully functional accommodation booking platform similar to Airbnb using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The system is designed to provide a seamless digital environment where travelers can easily discover and book properties while hosts can list and manage their accommodations efficiently.

The platform integrates several essential features such as secure user authentication, responsive design, property management, search and filtering, AI-based recommendations, analytics dashboards, and third-party API integrations. Additional functionalities like PDF booking confirmations, notifications, user profiles, blogs, and social media integration further enhance the overall usability and user experience.